

Multivariate Analysis: Assessment 4

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Question 1

Suppose $X \sim N_p(\mu, \Sigma)$. Show that the maximum likelihood estimator of Σ is biased, and give the bias.

Solution

Let $X_1, X_2, \dots, X_n$ be a random sample from a p -dimensional multivariate normal distribution $N_p(\mu, \Sigma)$.

The likelihood function for the sample is:

$$L(\mu, \Sigma) = \prod_{i=1}^n \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (X_i - \mu)' \Sigma^{-1} (X_i - \mu) \right\}$$
$$L(\mu, \Sigma) = (2\pi)^{-np/2} |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n (X_i - \mu)' \Sigma^{-1} (X_i - \mu) \right\}$$

The maximum likelihood estimators obtained by maximizing the log-likelihood function:

$$\ell(\mu, \Sigma) = -\frac{np}{2} \ln(2\pi) - \frac{n}{2} \ln |\Sigma| - \frac{1}{2} \sum_{i=1}^n (X_i - \mu)' \Sigma^{-1} (X_i - \mu)$$

Taking derivatives and setting to zero yields:

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'$$

To find the expected value of $\hat{\Sigma}$, we use the following decomposition:

$$\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' = \sum_{i=1}^n (X_i - \mu)(X_i - \mu)' - n(\bar{X} - \mu)(\bar{X} - \mu)'$$

Taking expectations:

$$E \left[\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \right] = \sum_{i=1}^n E[(X_i - \mu)(X_i - \mu)'] - nE[(\bar{X} - \mu)(\bar{X} - \mu)']$$

Now, we know that: - For each i , $E[(X_i - \mu)(X_i - \mu)'] = \Sigma$ since $\text{Var}(X_i) = \Sigma$ - For the sample mean, $\text{Var}(\bar{X}) = \frac{\Sigma}{n}$, therefore:

$$E[(\bar{X} - \mu)(\bar{X} - \mu)'] = \frac{\Sigma}{n}$$

Substituting these results:

$$\begin{aligned} E \left[\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \right] &= \sum_{i=1}^n \Sigma - n \left(\frac{\Sigma}{n} \right) \\ E \left[\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \right] &= n\Sigma - \Sigma = (n-1)\Sigma \end{aligned}$$

Now, the expectation of $\hat{\Sigma}$ is:

$$\begin{aligned} E[\hat{\Sigma}] &= E \left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \right] \\ &= \frac{1}{n} E \left[\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \right] \\ &= \frac{1}{n} \cdot (n-1)\Sigma = \frac{n-1}{n} \Sigma \end{aligned}$$

Therefore, the bias is:

$$\begin{aligned}
\text{Bias}(\hat{\Sigma}) &= E[\hat{\Sigma}] - \Sigma \\
&= \frac{n-1}{n} \Sigma - \Sigma \\
&= -\frac{1}{n} \Sigma
\end{aligned}$$

The maximum likelihood estimator $\hat{\Sigma}$ is biased downward, with bias $-\frac{1}{n}\Sigma$. This means the MLE systematically underestimates the true covariance matrix. The unbiased estimator is given by:

$$S = \frac{n}{n-1} \hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'$$

which satisfies $E[S] = \Sigma$.

Question 2

Marginal normality

In Figure 1, all variables appear to be marginally normally distributed, except for a possible outlier in the max breadth plot. After conducting a test for normality, the large p-values in Table 1 indicate that bas height, bas length and nasal height do not deviate significantly from normality. Max breadth has a p-value of 0.04, which is much lower than the other explanatory variables. A p-value of 0.04 provides some evidence against normality. However, the Shapiro-Wilk test is sensitive to outliers and the outlier seen in Figure 1 may be causing this low p-value. Visually, max breadth appears to be normally distributed.

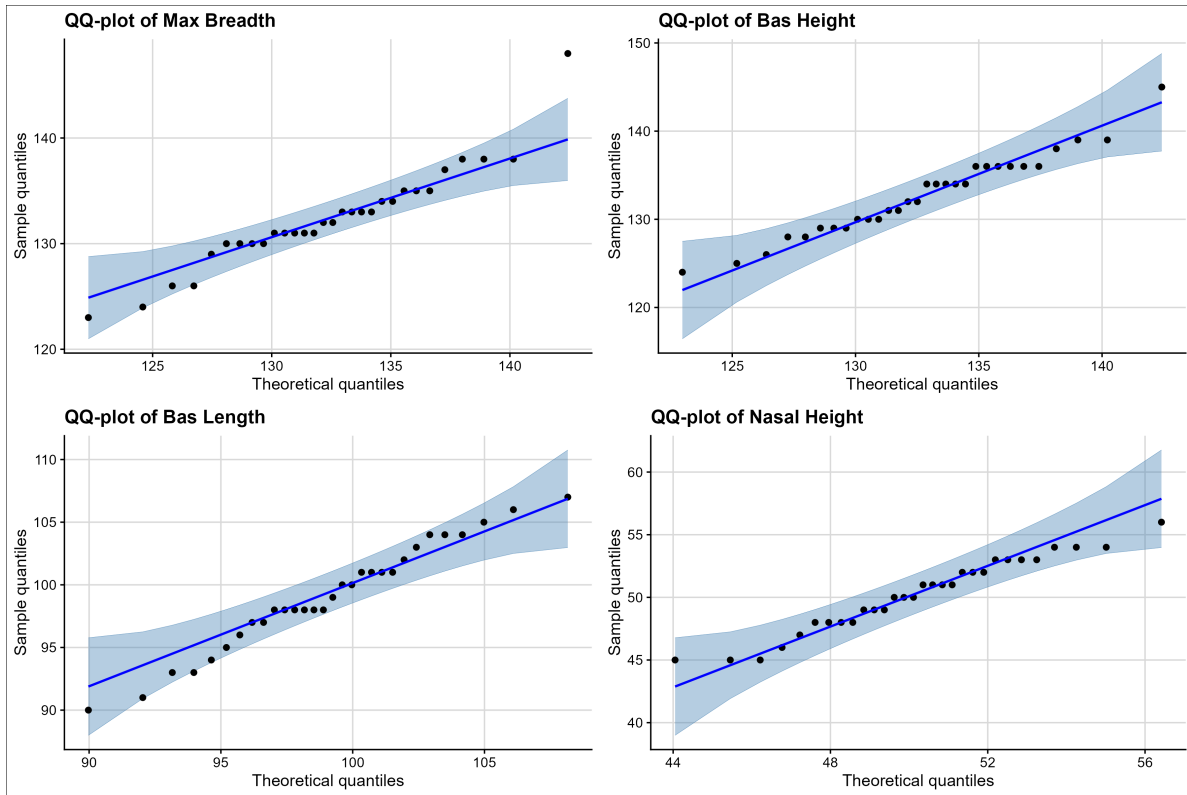


Figure 1: QQ-plots to assess marginal normality for maximal breadth of skull, basibregmatic height of skull, basialveolar length of skull and nasal height of skull.

Table 1: Shapiro-Wilk test for normality.

	p-value
Maximal breadth of skull	0.0403
Basibregmatic height of skull	0.5799
Basialveolar length of skull	0.7495
Nasal height of skull	0.3762

Multivariate normality

13.33 % of Mahalanobis distances are less than the chi-square median, 1.3863. If the data was multivariate normally distributed, we would expect that approximately 50% of distances would be less than the critical value. This indicates a deviation from multivariate normality. As seen in Figure 2 the larger distances have the most deviation from the chi-square median value.

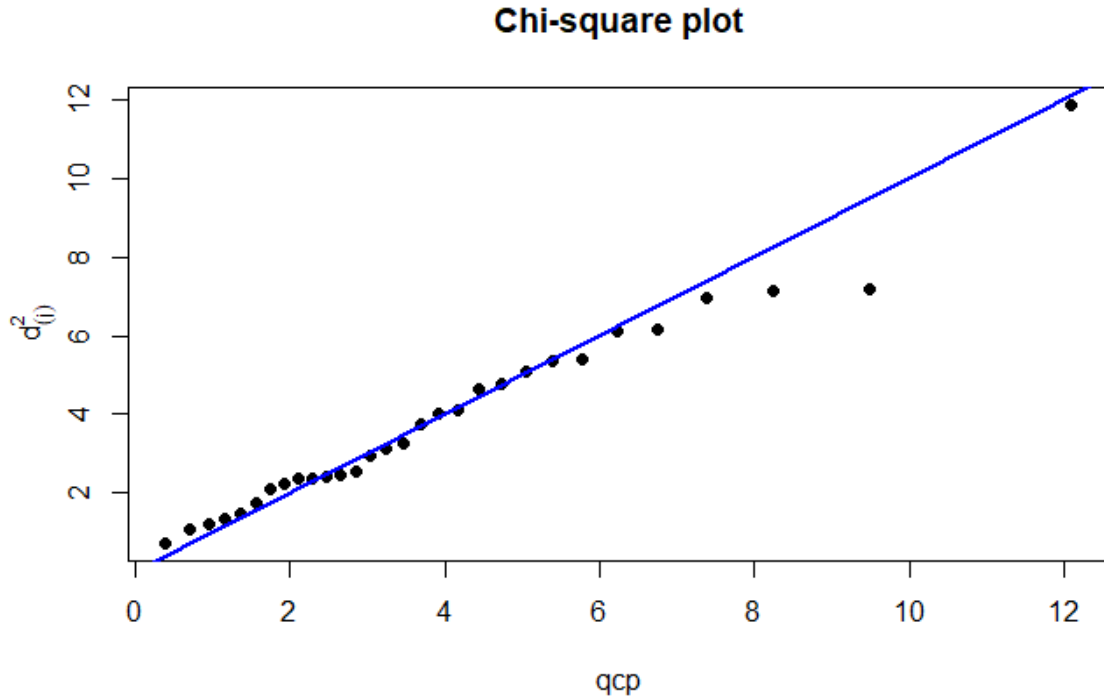


Figure 2: Chi-square plot of Mahalanobis distances to evaluate multivariate normality of maximal breadth of skull, basibregmatic height of skull, basialveolar length of skull and nasal height of skull.